



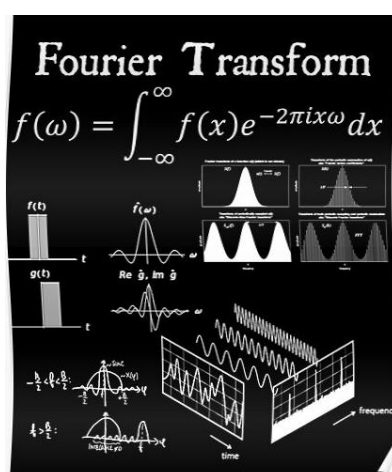
REDHILL SCHOOL



Tour de Maths 2023

Leg 5

Question Paper



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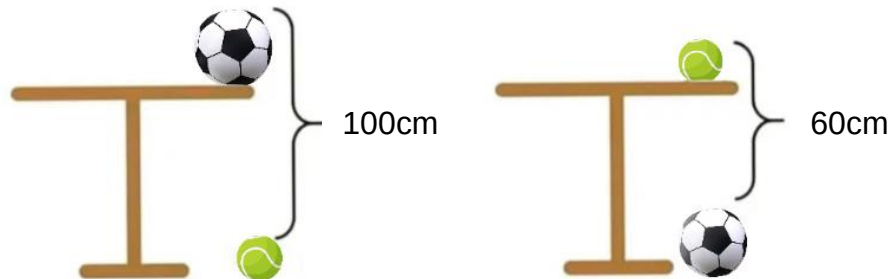


PRESH TALWALKAR

5 MARK QUESTIONS

QUESTION 1

What is the height of the table shown in the image below?



QUESTION 2

If $a \oplus b = ab + \frac{a}{b}$, determine the value of $6 \oplus 2$.

QUESTION 3

Find the number formed by the last three digits of the sum

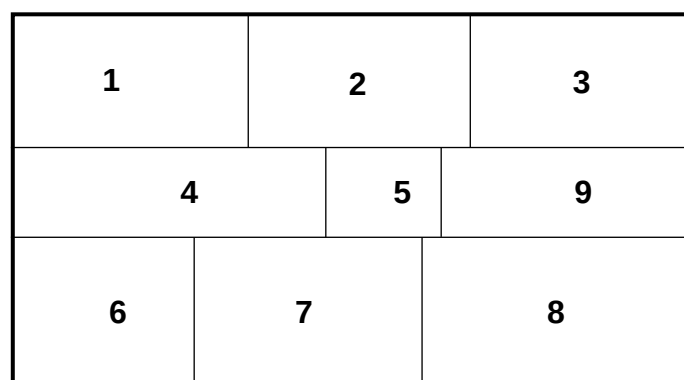
$$2! + 4! + 6! + 8! + \dots + 2018!$$



Where for a positive integer n , $n!$ is " n factorial" which is the product of the integers from 1 to n , i.e. $1 \times 2 \times 3 \times 4 \times \dots \times n$.

QUESTION 4

In the diagram, a large rectangle is divided into nine smaller rectangles which are labelled 1 to 9, as shown below.



In how many ways can exactly two of these nine rectangles be shaded so that the shaded rectangles are not touching?

QUESTION 5

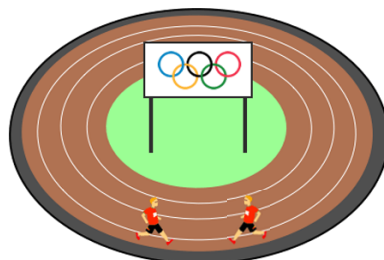
In how many ways can 22 be written as the sum of 3 different prime numbers? That is, determine the number of triples $(a; b; c)$ of prime numbers with $1 < a < b < c < 23$ and $a + b + c = 22$.

QUESTION 6

Two people, starting at the same point, are running in opposite directions around a 400-metre circular path for 10 minutes.

If the speed of person A is 4 metres per second and the speed of person B is 5 metres per second.

How many times will person A and B pass each other?



QUESTION 7

How many numbers are there in the list 1; 2; 3; 4;; 10000 which contain exactly two consecutive 9's such as 993, 1992 and 9929 but not 9295 or 1999?

QUESTION 8

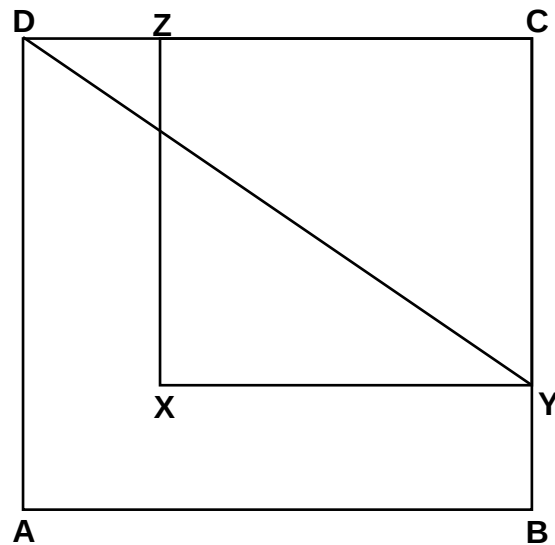
Suppose $X = 1 + 11 + 101 + 1001 + 10001 + \dots + \overbrace{1000\dots0001}^{50 \text{ ZEROES}}$

When X is calculated and written as a single integer, the sum of its digits is?

- (A) 58 (B) 99 (C) 55 (D) 50 (E) 103
-

QUESTION 9

ABCD and XYCZ are squares. The area inside square ABCD, but outside square XYCZ is 30 square units.



Determine the length of CD if $DY = 10$ units.

QUESTION 10

What is the unit digit of $5^{100} - 2^{100}$?

- (A) 5 (B) 0 (C) 2 (D) 4 (E) 9
-

10 MARK QUESTIONS

QUESTION 11

A certain school is throwing a dance. The student council wants to give a welcome gift to each student that attends the dance.

Gifts-R-Us charges R1,00 per gift. However, if they were to purchase the gifts at *Gifts-R-Us*, they would exceed their budget by R17.

At *Presents-4-U*, they only charge R0,80 per gift. At this price, the student council would have R5 left over in their budget.

Determine the number of gifts the student council is planning to buy.

QUESTION 12

If it is given that $a+1=b+2=c+3=d+4=a+b+c+d+5$, what is the value of $a+b+c+d$?

QUESTION 13

Three, fair, six-sided dice are thrown.



What is the probability that the sum of the three dice is 9?

QUESTION 14

A number is called special if it is the sum of a non-negative multiple of 4 and a non-negative multiple of 7.

So, 11 and 12 are special since $11 = 1 \times 4 + 1 \times 7$ and $12 = 3 \times 4 + 0 \times 7$ but 13 is not special.

How many special numbers are there in the interval 1 to 1000?

QUESTION 15

In Miss Q's class $\frac{1}{3}$ of the students brought a teddy bear to school.

Each boy in the class took 12 books, each girl took 17 books, and each teddy bear took 9 books.



A total of 305 books were taken out.

How many girls were there in Miss Q's class?

20 MARK QUESTIONS

QUESTION 16

An arithmetic (also known as linear) sequence is a sequence in which each term after the first is obtained by adding a constant to the previous term.

For example: 4, 7, 10, 13, 16 is an arithmetic sequence with five terms, in which the first term is 4 and the constant added is 3.

Each row and each column in the 5 x 5 array below is an arithmetic sequence with five terms.

1				17
	A			
25				81

Calculate the value of A.

QUESTION 17

In the diagram below, a positive integer is hidden behind each star.

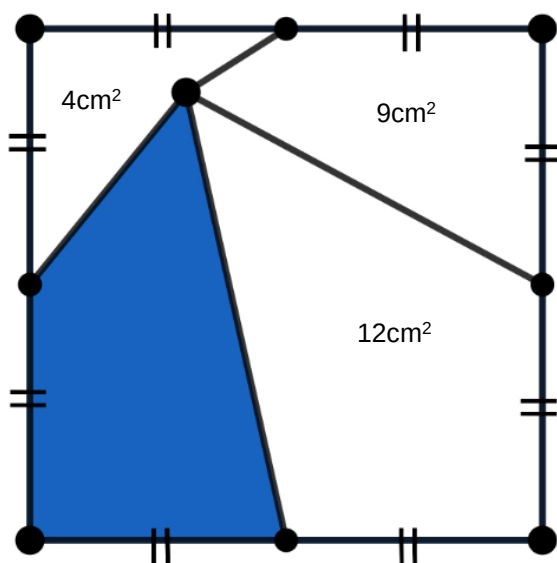
The integer shown on each star is the product of the integers hidden behind the other three stars.



What is the product of all four hidden integers?

QUESTION 18

Determine the area of the blue shaded region below.



QUESTION 19

Twelve squares are placed in a row forming the grid below. Each square is to be filled with an integer. After the third square, each integer in a square is the sum of the previous three integers. If we know the third integer is 6, the sixth integer is 11, and the eleventh integer is 14.

		6		Q	11					14	
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Determine the value of Q .

QUESTION 20

Noah and Acacia each have nine cards, numbered from 1 to 9. Each person randomly discards one of their cards and then calculates the sum of the numbers on their remaining eight cards.

Calculate the probability that the difference between these two sums is a multiple of 4.

